WORKED SOLUTION

STEP I 1989, Question 2

Pure • Riemann Sums and Limits

STEP I 1989, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q2

The key idea

The strange expression is nothing more than a sum of rectangle areas in disguise. If we write down the total area of the n rectangles directly, we get a sum involving $k \ln k$; the factorial logarithms are just that same sum regrouped. Once the two are matched up, the limit is the integral, which the first part lets us evaluate.

Solution

The indefinite integral

Integrating by parts with $u = \ln x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = x$, so that $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$ and $v = \frac{1}{2}x^2$,

$$\begin{aligned}\int x \ln x \,\mathrm{d}x &= \tfrac{1}{2}x^2 \ln x - \int \tfrac{1}{2}x^2 \cdot \frac{1}{x}\,\mathrm{d}x \\ &= \tfrac{1}{2}x^2 \ln x - \tfrac{1}{4}x^2 + C.\end{aligned}$$

The value of the integral

Since $\ln(1/x) = -\ln x$, the integrand is $-x \ln x$. The integrand isn't defined at $x = 0$, so we integrate over $[\varepsilon, 1]$ and let $\varepsilon \to 0^+$:

$$\int_\varepsilon^1 x \ln(1/x)\,\mathrm{d}x = \left[\tfrac{1}{4}x^2 - \tfrac{1}{2}x^2 \ln x\right]_\varepsilon^1 = \tfrac{1}{4} - \tfrac{1}{4}\varepsilon^2 + \tfrac{1}{2}\varepsilon\,(\varepsilon \ln \varepsilon).$$

As $\varepsilon \to 0$, $\varepsilon^2 \to 0$ and $\varepsilon \ln \varepsilon \to 0$ (the given fact), so

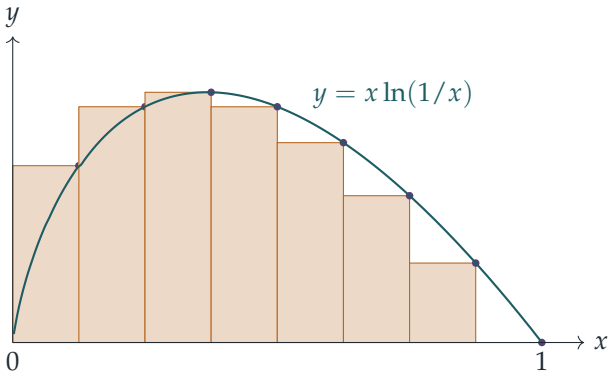
$$\int_0^1 x \ln(1/x)\,\mathrm{d}x = \frac{1}{4}.$$

The same fact shows that $f(x) = x \ln(1/x)$ tends to 0 as $x \to 0^+$, so setting $f(0) = 0$ makes f continuous on the whole of $[0, 1]$. The curve rises from 0 to a maximum of $\frac{1}{\mathrm{e}}$ at $x = \frac{1}{\mathrm{e}}$ (where $f'(x) = -\ln x - 1 = 0$) and falls back to 0 at $x = 1$.

The rectangles

Split $[0,1]$ into n strips of width $\frac{1}{n}$. The kth strip is $[\frac{k-1}{n}, \frac{k}{n}]$, and its rectangle has its top right-hand corner on the curve, so its height is

$$f\left(\frac{k}{n}\right) = \frac{k}{n}\ln\frac{n}{k} = \frac{k}{n}(\ln n - \ln k).$$



The total area of the rectangles is therefore

$$\begin{aligned} S_n = \sum_{k=1}^{n} \frac{1}{n}\cdot\frac{k}{n}(\ln n - \ln k) &= \frac{\ln n}{n^2}\sum_{k=1}^{n} k - \frac{1}{n^2}\sum_{k=1}^{n} k\ln k \\ &= \frac{\ln n}{n^2}\cdot\frac{n(n+1)}{2} - \frac{1}{n^2}\sum_{k=1}^{n} k\ln k \\ &= \frac{1}{2}\left(1+\frac{1}{n}\right)\ln n - \frac{1}{n^2}\sum_{k=1}^{n} k\ln k. \end{aligned}$$

The first term already matches the expression we want.

Ask yourself Where do the factorials come from?

For $0 \le m \le n-1$,

$$\ln\frac{n!}{m!} = \ln((m+1)(m+2)\cdots n) = \sum_{k=m+1}^{n} \ln k.$$

Now add these up for $m = 0, 1, \ldots, n-1$ and count how often each $\ln k$ appears. The term $\ln k$ is included exactly when $m+1 \le k$, that is for $m = 0, 1, \ldots, k-1$: that's k times. Hence

$$\ln\frac{n!}{0!} + \ln\frac{n!}{1!} + \cdots + \ln\frac{n!}{(n-1)!} = \sum_{k=1}^{n} k\ln k.$$

(For example, with $n = 3$: $\ln 6 + \ln 6 + \ln 3 = \ln 1 + 2\ln 2 + 3\ln 3$, both equal to $\ln 108$.) So the expression in the question is exactly S_n, the area of the n rectangles.

The limit

As $n \to \infty$ the rectangles fit the curve ever more closely, and their total area tends to the area under the curve, since f is continuous on $[0, 1]$. Therefore

$$\frac{1}{2}\left(1+\frac{1}{n}\right)\ln n - \frac{1}{n^2}\left[\ln\frac{n!}{0!} + \ln\frac{n!}{1!} + \cdots + \ln\frac{n!}{(n-1)!}\right] = S_n \to \int_0^1 x\ln(1/x)\,\mathrm{d}x = \frac{1}{4},$$

as required.

If you want something firmer than "the rectangles fit ever more closely", here is a bound. On each strip, the rectangle and the area under the curve differ by at most $\frac{1}{n}$ times the amount by which f varies across that strip. Add these variations over all the strips: f climbs by $\frac{1}{\mathrm{e}}$ and comes back down by $\frac{1}{\mathrm{e}}$, so the total is at most $\frac{2}{\mathrm{e}}$. Hence $|S_n - \frac{1}{4}| \le \frac{2}{\mathrm{e}n} \to 0$.

Checks

- Differentiate the answer to the first part: $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}x^2\ln x - \frac{1}{4}x^2\right) = x\ln x + \frac{1}{2}x - \frac{1}{2}x = x\ln x$.
- With $n = 2$ the expression is $\frac{3}{4}\ln 2 - \frac{1}{4}(\ln 2 + \ln 2) = \frac{1}{4}\ln 2 \approx 0.173$, which is the area of the two rectangles, $\frac{1}{2}\cdot\frac{1}{2}\ln 2 + \frac{1}{2}\cdot 0$.
- Numerically, the expression is about 0.24559 for $n = 10$, 0.249937 for $n = 100$ and 0.2499992 for $n = 1000$, closing in on $\frac{1}{4}$.

Common slips

- Losing the sign: the integrand is $x\ln(1/x) = -x\ln x$, so the antiderivative from the first part must be negated, and the area is $+\frac{1}{4}$, not $-\frac{1}{4}$.
- Writing $\sum k = \frac{1}{2}n^2$ instead of $\frac{1}{2}n(n+1)$: the $\frac{1}{n}$ inside the bracket would then go missing.
- Forgetting that the integral is improper at 0: the lower limit needs the fact $x\ln x \to 0$, which the question gives you.

Beyond STEP

The rectangle sum is a Riemann sum, and the fact that it converges to the integral for every continuous function on a closed interval is one of the first big theorems of a university analysis course. The bound we used, with the total rise and fall of f, is the idea of a function of *bounded variation*. Turned round, this question also gives information about $\sum k\ln k$, a quantity closely related to the hyperfactorial $1^1\,2^2\cdots n^n$, whose growth can be found more precisely in the same spirit as Stirling's formula for $n!$. Abbott's *Understanding Analysis* develops the Riemann integral carefully.

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